

MOBILE SOFT PLC SIMULATOR

5-Year Growth Strategy Report

Revenue Forecasts · Feature Roadmap · Partnership Strategy

July 2026 | Confidential — Prepared for Jarrod Franz

\$882K 5-Year Base Revenue Conservative–Optimistic range	2,410 Base Paying Users (Y5) Across all tiers	\$376K Year 5 ARR (Base) 706% over 5 years
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This report extends the Year-1 Business Analysis with a five-year compounding revenue model across three scenarios, a prioritised feature roadmap grounded in the current Phase 14 codebase, and a detailed partnership strategy targeting hardware vendors, SCADA platforms, and educational institutions.

1. Executive Summary

Mobile Soft PLC Simulator enters the market as the world's first fully in-app, cross-platform industrial PLC simulator — shipping all five IEC 61131-3 languages, four live protocol servers (OPC UA, Modbus TCP, MQTT + Sparkplug B, DNP3), and an advanced process simulation engine, all without requiring a companion process or cloud dependency. This is a rare combination of technical depth and accessibility that no competing product in the sub-\$200 price range replicates.

The five-year revenue model presented here is built on three compounding growth scenarios. The base case reaches \$375,829 in Year 5 and \$882,478 cumulative — driven by organic subscriber growth (approx. 25% year-on-year), the progressive addition of education site licences from Year 2, and partnership/hardware revenue entering from Year 3. The optimistic case exceeds \$1.14 million cumulative if a B2B education or SCADA-platform integration deal lands in Year 2–3. The conservative case, representing slower organic discovery without partnership revenue, still reaches \$382,000 over five years — comfortably profitable for a solo/small-team product with near-zero marginal cost.

Feature investment is prioritised around three themes: PLCopen XML interoperability (the most-cited missing capability for integrators), AI-assisted code generation (a differentiation opportunity before a competitor arrives), and enterprise multi-seat management (the unlock for institutional and B2B bulk purchasing). Partnership strategy targets Inductive Automation as the highest-leverage first mover — the OPC UA integration is already confirmed working end-to-end with Ignition 8.3, giving an immediate co-marketing hook — followed by Arduino/Opta for hardware bundles and vocational training institutions for site licensing.

✓ All revenue figures are gross. Apple App Store and Google Play each retain 15% commission (Small Business Program) on mobile revenue. Windows direct sales incur only ~2–3% payment processing fees. Education and partnership revenue is direct. Net margin after platform fees and estimated operating costs is modelled at Section 3.

2. Five-Year Revenue Model

2.1 Revenue Streams

The model comprises five revenue streams that activate progressively across the five-year horizon:

- **Mobile Monthly Pro (\$12.99/month):** Subscriber base grows via App Store/Play Store organic discovery, content marketing, and community growth. Churn of 4% per month assumed (B2B app benchmark for SMB). Subscribers carry forward month-to-month.
- **Mobile Annual Pro (\$89.99/year):** Annual subscribers commit upfront. Year 2 onward includes 80% renewal rate on prior-year annual cohort, compounding the revenue floor significantly.
- **Windows Perpetual Licence (\$129.99 direct):** One-time purchase via direct website download (0% commission). Sold to desktop-centric engineers, IT departments, and SCADA integrators who prefer a standalone installed licence.
- **Education Site Licences (\$599–\$1,499/institution/year):** Semester-length site licences for TAFE colleges, community colleges, and university departments. Begins Year 2 via outreach; scales with partnership deals from Year 3. Averaged at \$800/institution for modelling.
- **Partnership & Hardware Revenue:** Distribution commissions, bundled hardware deals (Arduino Opta, RPi I/O boards), white-label licensing, and API-tier access. Begins conservatively in Year 3 once partnerships are formalised.

2.2 Pricing Architecture

Tier	Price	Billing	Commission	Net to Developer
Mobile Monthly Pro	\$12.99/month	Monthly auto-renew	15% (Apple/Google)	\$11.04/sub/month
Mobile Annual Pro	\$89.99/year	Annual auto-renew	15% (Apple/Google)	\$76.49/sub/year
Windows Perpetual	\$129.99 once	One-time	~2.5% (Stripe)	\$126.74/sale
Education Site Licence	\$599–\$1,499/yr	Annual direct invoice	0% (direct)	\$599–\$1,499/yr
AI/Enterprise Add-on	\$29.99/month	Monthly (future)	15%	\$25.49/sub/month

2.3 Growth Assumptions by Scenario

Three scenarios are modelled. All share the same Year-1 launch trajectory derived from App Store benchmarks (9.1% download-to-trial conversion, 35–42% trial-to-paid, ~\$35 RLTV per payer at launch). Divergence accelerates from Year 2 based on marketing investment, partnership success, and feature velocity.

Assumption	Conservative	Base	Optimistic
Organic sub growth (Y/Y)	12–15%	22–27%	38–45%
Monthly churn rate	5.5%	4.0%	2.5%
Annual renewal rate	72%	80%	88%

Education deals (Y2-Y5)	2 / 5 / 10 / 18	5 / 15 / 30 / 55	10 / 30 / 70 / 130
Partnership revenue starts	Year 4	Year 3	Year 2
Windows sales growth (Y/Y)	15%	22%	35%
AI/Enterprise tier launches	Not modelled	Year 4	Year 3

2.4 Year-by-Year Revenue Breakdown — Base Case

The base case assumes consistent execution: app store launch in Q1 Year 1, active community building from Month 6, first education partnership in Year 2, and hardware/ecosystem partnerships from Year 3 onward. Growth decelerates from high early rates as the subscriber base matures but is sustained by new revenue streams.

Revenue Stream	Year 1	Year 2	Year 3	Year 4	Year 5
Mobile Monthly Pro	\$15,588	\$23,382	\$34,295	\$51,443	\$77,940
Mobile Annual Pro	\$17,998	\$25,197	\$35,996	\$50,394	\$70,192
Windows Perpetual	\$12,999	\$16,899	\$22,099	\$28,598	\$37,697
Education Site Licences	\$0	\$12,000	\$30,000	\$60,000	\$100,000
Partnership & Hardware	\$0	\$5,000	\$20,000	\$45,000	\$90,000
TOTAL GROSS REVENUE	\$46,585	\$82,478	\$142,390	\$235,435	\$375,829
Platform Fees (est. 10%)	-\$3,459	-\$4,558	-\$5,626	-\$7,843	-\$11,214
NET REVENUE (after fees)	\$43,126	\$77,920	\$136,764	\$227,592	\$364,615
YoY Growth	—	+77%	+73%	+65%	+60%

2.5 Subscriber Base Growth — Base Case

User Segment	Y1 End	Y2 End	Y3 End	Y4 End	Y5 End
Monthly Pro (active subs)	150	225	340	510	765
Annual Pro (active subs)	200	340	510	715	1,000
Windows (cumulative licences)	100	180	285	430	645
Education Institutions	0	15	38	75	125
Total Paying Users	450	745	1,135	1,655	2,410

2.6 Three-Scenario Revenue Summary

The following table compares all three scenarios at an annual level. Note that Year-1 figures are the same across all scenarios (launch year is identical); scenarios diverge from Year 2 based on the assumptions in Section 2.3.

Year	Conservative	Base Case	Optimistic
Year 1	\$46,585	\$46,585	\$46,585
Year 2	\$57,200	\$82,478	\$95,400
Year 3	\$72,000	\$142,390	\$175,000
Year 4	\$91,500	\$235,435	\$305,000
Year 5	\$115,000	\$375,829	\$520,000
5-Year Total	\$382,285	\$882,717	\$1,141,985
Y5 Annual Run Rate	\$115K	\$376K	\$520K
Implied ARR Multiple (4.6×)	\$529K	\$1.73M	\$2.39M

 Implied valuation uses the 4.6× ARR multiple benchmark for niche vertical SaaS products observed in 2025–2026 (Carta / SaaS Capital data). This is an indicative exit valuation for an acquisition, not a guaranteed outcome. A bootstrapped solo-founder product at \$376K ARR would likely attract strategic acquirer interest from industrial software consolidators.

2.7 Cost Structure & Profitability

Mobile Soft PLC carries near-zero marginal cost of delivery — no cloud infrastructure required for the core product (all logic runs on-device). Operating costs are primarily the developer's time, app store fees, and minimal SaaS tooling.

Cost Category	Year 1	Year 2	Year 3	Year 4	Year 5
Apple/Google/Stripe fees	\$3,459	\$4,558	\$5,626	\$7,843	\$11,214
App Store membership / tools	\$1,200	\$1,200	\$1,400	\$1,600	\$2,000
Marketing / content / ads	\$3,000	\$5,000	\$8,000	\$12,000	\$18,000
Cloud infra (future sync/API)	\$0	\$1,200	\$2,400	\$4,000	\$6,000
Legal / accounting	\$1,500	\$1,500	\$2,000	\$2,500	\$3,000
Total Operating Costs	\$9,159	\$13,458	\$19,426	\$27,943	\$40,214
Net Profit (Base Case)	\$37,426	\$69,020	\$122,964	\$207,492	\$335,615
Net Profit Margin	80%	84%	86%	88%	89%

 By Year 3, net profit exceeds \$120K at an 86% margin — placing this product firmly in the "high-quality lifestyle business" quadrant. Year 4–5 margins approach 89%, consistent with mature niche SaaS benchmarks.

3. Feature Additions to Boost Value & Revenue

Mobile Soft PLC Simulator is technically advanced at Phase 14 — all five IEC 61131-3 languages execute, four industrial protocols are live in-app, and the process simulation engine includes PID auto-tuning and MIMO models. The opportunity is not to add breadth indiscriminately; it is to close the four gaps that prospective buyers consistently cite as purchase blockers, then add capabilities that open entirely new market segments.

Features below are prioritised by revenue impact and development effort. Each card estimates the expected revenue lift (incremental ARR potential once active) and categorises development effort on a 1–5 scale.

3.1 Priority 1 — Purchase Blockers (Build First)

Priority 1 — High Impact Impact: Very High — unlocks integrators and converts "try" to "buy" | Effort: Medium (3–5 months)

PLCopen XML Import / Export

PLCopen XML (.xml) is the standard interchange format used by every major PLC IDE — Allen-Bradley RSLogix/Studio 5000, Siemens TIA Portal, CODESYS, Beckhoff TwinCAT, and more. The PROJECT_BRIEF explicitly lists this as "not yet implemented." For the app's primary users (integrators and engineers with existing programs), the inability to import their own projects is a top-3 objection. Implementing full export unlocks the app as a "test bed for real programs" — a capability no other mobile tool offers. Expected ARR lift: \$40K–\$80K from integrator conversions and education adoption.

Priority 1 — High Impact Impact: High — directly drives annual subscription retention | Effort: Low-Medium (4–8 weeks)

Cloud Project Sync & Backup (Google Drive / iCloud)

Cloud sync solves a significant friction point: engineers want their simulator projects on whichever device they pick up (tablet on the shop floor, laptop at the desk, phone while travelling). A native Drive/iCloud sync creates a habit loop that makes the annual subscription "sticky." It also functions as a backup and disaster recovery feature that supports selling to enterprises and institutions — both groups have policies requiring backup. Positioning: "Your logic programmes, accessible everywhere — no cable, no export, no re-entry." Expected ARR impact: primary driver of annual-to-monthly upgrade conversions and reduced churn. Estimated 15–20% reduction in annual churn.

Priority 1 — High Impact Impact: Very High — new revenue tier; major differentiator | Effort: High (4–6 months, requires LLM API integration)

AI-Assisted Code Generation & Fault Diagnostics

This is the highest-differentiation move available before a competitor with deeper resources adds it first. "Copilot for PLC engineers" — a system prompt-tuned LLM that understands IEC 61131-3 syntax, the app's tag schema, and industrial control conventions. Concrete features: natural-language-to-Ladder-Logic ("Write a TON timer rung that energises Pump_Run 5 seconds after Tank_High"), fault tree analysis ("Why is Conveyor_Fault latched?"), and code review ("Explain this ST function block"). Delivered as a new "AI Pro" tier at \$29.99/month (or \$199/year). Expected ARR lift at 10% of base paid subscribers upgrading: \$18K–\$45K annually.

3.2 Priority 2 — Strategic Revenue Unlocks (Year 2–3)

Priority 2 — Strategic Impact: Very High — opens B2B bulk purchasing | Effort: High (3–5 months)

Enterprise Multi-Seat Console & Team Licensing

A web-based admin console that allows a company, training institution, or integrator firm to provision, manage, and monitor multiple licensed seats from one dashboard. Features: bulk user provisioning, seat count management, usage analytics (which programmes, which protocols, scan times), and group project libraries (shared templates). Pricing: \$49/seat/year for 10–50 seats, \$35/seat/year for 50+ seats. This directly

targets the TAFE, community college, and corporate training segment. A single deal with a 20-seat institution at \$49/seat generates \$980/year — equivalent to 11 individual annual subscribers. Expected ARR lift: \$30K–\$90K annually by Year 3.

Priority 2 — Strategic Impact: High — unlocks European market and building automation vertical | Effort: Medium (per protocol 4–8 weeks)

S7comm & BACnet Protocols (Phase 14 Completions)

Phase 14 currently has EtherNet/IP shipped; S7comm (Siemens), BACnet, FINS (Omron), and SLMP (Mitsubishi) are pending. S7comm is the highest-priority completion: Siemens is the world's largest PLC vendor by installed base, and European engineers disproportionately use Siemens platforms. A functional S7comm outstation makes the app directly useful for Siemens SCADA/WinCC developers — a large, underserved segment of the app's target market. BACnet opens the building automation vertical (HVAC, lighting, access control) — a \$5.5B adjacent market with distinct personas (BAS technicians, commissioning engineers) who are currently not served by any mobile simulator. Expected ARR lift from S7comm alone: \$20K–\$50K annually via European App Store discovery.

Priority 2 — Strategic Impact: High — extends the "always-on" value proposition beyond the device | Effort: Medium-High (2–3 months)

Remote Live Dashboard (Browser-Accessible Monitoring)

A zero-configuration web dashboard that streams the running simulator's tag values to any browser on the local network via WebSocket. No app install required on the monitoring device. Use case: engineer runs simulator on their phone/tablet connected to SCADA client, supervisor monitors the dashboard on their PC without installing anything. This is a significant demonstration capability — at trade shows, in classrooms, in client presentations. It also lays the groundwork for the cloud-hosted monitoring tier (a future premium feature). Positioned as a Pro feature; adds tangible "wow factor" in sales demos.

3.3 Priority 3 — Curriculum & Community (Year 2–3)

Priority 3 — Mid-Term Impact: High for education segment — directly supports site licensing | Effort: Medium (3–4 months)

In-App Learning Paths & Certification Mode

A structured tutorial and quiz system built into the app, with progress tracking and a printable certificate of completion. Topics: Introduction to Ladder Logic, TON/TOF timers, PID control fundamentals, Modbus register mapping, OPC UA subscription setup. Partners with TAFE/college institutions who need measurable student outcomes for curriculum accreditation. Also creates a direct entry point for Udemy/Coursera integration — instructors can direct students to the app's certification mode as coursework. Positions the app as an education platform rather than just a tool, justifying institutional pricing and government training grant eligibility.

Priority 3 — Mid-Term Impact: Medium — drives discovery and trial-to-paid conversion | Effort: Low-Medium (6–8 weeks for core; ongoing for content)

Project Template Library & Community Marketplace

A curated library of pre-built project templates: conveyor sorter, tank level control, HVAC temperature loop, motor starter with interlocks, batch reactor sequence. Templates are immediately runnable and configurable — engineers can load a conveyor template and adapt it to their use case in minutes. Community contribution model: Pro users can publish templates; highly-rated templates are featured. This drives App Store search rankings ("PLC ladder logic examples"), improves trial engagement, and reduces time-to-value — the single most important metric for trial-to-paid conversion. Potential for paid template packs as a supplemental revenue stream (\$4.99 per domain-specific pack).

Priority 3 — Mid-Term Impact: Medium-High — creates tangible product and hardware partnership hook | Effort: High (4–6 months hardware + firmware)

Hardware I/O Companion Board (Raspberry Pi / ESP32)

A small companion hardware board — ESP32 or Raspberry Pi Pico W based — with physical I/O: 8 digital inputs (push buttons), 8 digital outputs (LEDs), 2 analog inputs (potentiometers), and a small OLED status display. Connects to the phone/tablet via USB or WiFi. The simulator treats the board as real I/O: pressing a physical button sets a digital input tag; a real LED lights when the simulator drives the coil. Price point: \$39–\$79 as a kit (BOM cost ~\$12–18 at volume). Positions Mobile Soft PLC as the "Arduino of PLC learning" — tangible hardware lowers the psychological barrier for students and hobbyists. See Section 4 for partnership options around this concept.

3.4 Priority 4 — Long-Term Differentiation (Year 3–5)

Priority 4 — Complementary Impact: Medium — B2B / integrator value-add | Effort: Low–Medium (4–6 weeks)

Automatic IEC 61131-3 Programme Documentation Generator

Generates a formatted PDF or Word document from an open project: variable list with types and descriptions, ladder rung-by-rung narrative, I/O assignment table, timer/counter configuration summary. Integrators and FAT (Factory Acceptance Testing) engineers routinely produce this documentation manually. Automating it removes a significant administrative burden and positions the app as part of the professional workflow rather than a training tool. Priced as a Pro feature or a small in-app purchase for export access.

Priority 4 — Complementary Impact: High for software development companies | Effort: High (4–6 months, cloud infrastructure required)

API Access Tier — Virtual PLC as a Service

A REST + WebSocket API that allows external software to spawn a virtual PLC instance, load a programme, inject tag values, and read outputs — all programmatically. Target customer: test automation engineers and CI/CD pipelines for SCADA software companies. A SCADA vendor (Ignition, Kepware, ICONICS) could integrate a headless Mobile Soft PLC instance into their automated regression test suite. Pricing: \$299/month per API seat or usage-based. This is a platform-level move that requires cloud infrastructure but transforms the product from a simulator to an industrial testing platform — significantly elevating the category and valuation multiple.

4. Partnership Strategy

The strongest growth accelerator for a niche B2B tool is not advertising — it is being embedded in the workflow of organisations that already serve your target users. Each partnership category below is ranked by expected revenue leverage and actionability within a 12-month window.

4.1 Tier 1 — Immediate Opportunities (Act in Year 1)

Inductive Automation (Ignition SCADA)

Status: Ready to approach. The app's OPC UA server is already end-to-end verified with Ignition 8.3 — Browse, Read, Write, and Subscriptions all confirmed. This is the key technical proof required for any co-marketing conversation.

Ignition has over 60,000 active installations worldwide and a certified developer community of 12,000+ engineers. Inductive Automation runs a free Integrator Partner Program that lists compatible tools in their ecosystem directory. Listing Mobile Soft PLC as a "compatible OPC UA simulation device" in the Ignition ecosystem would immediately surface the app to every engineer and developer searching for a PLC to test against.

Approach: Join the Inductive Automation Integrator Partner Program (free). Submit a compatibility listing with a demo video showing a Mobile Soft PLC tag being subscribed to by Ignition. Propose a co-blog post or webinar to Ignition's developer community.

Revenue expectation: No direct revenue share, but 1,000+ Ignition-community-driven installs in Year 1-2, converting at standard trial-to-paid rates adds an estimated \$15K-\$40K ARR to the base case.

Arduino / Arduino Pro (PLC Opta Partnership)

Status: High-value approach; requires outreach. Arduino launched the PLC Opta in 2023 — a compact industrial PLC hardware at \$385 for the starter kit, targeting engineering education and industrial prototyping. Arduino also launched the Arduino PLC IDE, a desktop application that compiles IEC 61131-3 programs for the Opta hardware.

Mobile Soft PLC occupies a complementary position: "Can't afford the Opta hardware? Simulate the same logic on your phone for free." The two products share identical target users. A bundle partnership — where the Arduino Opta starter kit website prominently recommends Mobile Soft PLC for simulation — creates a no-cost referral channel into Arduino's 30-million-person community.

Partnership structure options: (a) Cross-promotional listing — Arduino's website links to the app, app links to Opta hardware; zero revenue share; (b) Bundle discount — App annual subscription included with Opta kit purchase at \$10/unit royalty; (c) Hardware I/O integration — Mobile Soft PLC natively recognises the Opta's I/O over USB, making the two products technically complementary. Option (c) is the highest value but requires firmware work.

Revenue expectation: A listing on Arduino's store page (30M+ visitors/month) could drive 200–500 qualified trial signups per month at a 35% conversion rate, representing \$8,000–\$22,000 additional MRR by Year 2.

4.2 Tier 2 — Strategic Education Partnerships (Year 1-2)

TAFE NSW / Australian Vocational Training Institutions

TAFE NSW delivers electrical and instrumentation training to thousands of apprentices and tradespersons annually. Their Certificate III/IV in Instrumentation and Control and Diploma of Engineering qualifications include PLC programming modules that currently rely on expensive Siemens S7 training benches or Allen-Bradley MicroLogix hardware (typically \$2,000–\$8,000 per training station).

A semester-length site licence at \$599–\$999 per institution per year is a compelling value proposition — it replaces one hardware training station and allows unlimited student access on personal devices. Target: 20–50 TAFE campuses in Years 2–3.

Approach: Contact the TAFE NSW Electrical and Instrumentation Department directly. Offer a free pilot semester licence in exchange for structured student feedback and a case study testimonial. A single positive case study from a recognised training institution is the most powerful B2B marketing asset available.

Udemy / Coursera — Course Integration

PLC programming courses are among the best-selling engineering categories on Udemy. Courses by instructors such as "RealPars" and "Garry Shortt" each have 50,000–150,000 enrolled students. These instructors currently recommend students purchase hardware (\$300–\$2,000) or use limited trial-ware from Siemens/Rockwell.

Partnership model: Negotiate with top-5 Udemy PLC course instructors to recommend Mobile Soft PLC as the "required practice tool" for their course. Provide a discounted instructor bundle code (\$29.99/student for annual Pro access vs. \$89.99 retail) with a 15% revenue share to the instructor. At 1,000 students/course, a single course partnership generates \$25,500 per cohort launch.

Udemy also has a direct "Udemy Business" content partnership track — content created around the app (tutorial videos embedded in the app or on YouTube) can be submitted as structured learning content.

Festo Didactic / SMC Education

Festo Didactic and SMC are the two dominant industrial training equipment vendors globally. Their training cells (electropneumatic systems, conveyor training benches, hydraulic trainers) are installed in engineering schools across 150+ countries. Both companies publish recommended software lists for accompanying their hardware.

Opportunity: Mobile Soft PLC can simulate the control logic that would drive a Festo Didactic training cell — the same Boolean logic, timer rungs, and I/O structure. Propose co-marketing: Festo recommends the app for students who need to pre-study or practice without physical lab access. Festo gains a zero-cost digital companion for their hardware; the app gains access to Festo's global reseller and academic network.

4.3 Tier 3 — Protocol & Standards Partnerships (Year 2-3)

OPC Foundation — Compliance Certification

The OPC Foundation operates a Compliance Testing Tool (CTT) program and maintains a certified-product registry at opcfoundation.org. Getting listed as a "Certified OPC UA Server" is the industrial equivalent of a

security certification — it immediately signals to enterprise procurement teams that the product has been independently validated.

The CTT test suite is freely downloadable. If the app's OPC UA implementation passes the relevant test cases (DA Server, Subscriptions, Security), Jarrod can apply for a compliance certificate and listing. This is a Year 2 initiative — it requires completing the OPC UA security hardening branch (feat/opcua-hardening per ARCHITECTURE.md) first. Cost: approximately \$5,000–\$15,000 for the formal certification process. Return: instant credibility with enterprise SCADA procurement teams.

ICS / SCADA Cybersecurity Training Firms

ICS cybersecurity is a fast-growing market (projected \$22.8B by 2028 at 9.8% CAGR). Training firms — Dragos, Claroty, Nozomi Networks, and specialised training providers like Idaho National Laboratory's NEET program — need safe, legal OT environments for trainee ethical hackers to practice against. Mobile Soft PLC, with its OPC UA, Modbus TCP, DNP3, and MQTT servers, is a ready-made "safe OT device" that can be attacked, probed, and analysed without any real-world risk.

Partnership model: License the app as a "virtual OT target device" to cybersecurity training providers. A site licence for a cybersecurity course provider at \$2,499/year for unlimited student sessions is a compelling price compared to the cost of maintaining physical PLC hardware labs. This is a small-volume, high-ACV (average contract value) segment that doesn't require high subscriber numbers to generate meaningful revenue.

AWS IoT / Azure IoT Hub — Cloud MQTT Integration

Mobile Soft PLC's MQTT + Sparkplug B adapter can already publish tag values to any broker. A formal integration guide and one-click configuration for AWS IoT Core, Azure IoT Hub, and Google Cloud IoT positions the app as an "IIoT edge device emulator" — a valuable developer tool for cloud IoT application developers who need a realistic OT data source without deploying physical hardware.

This opens distribution via AWS Marketplace and Azure Marketplace listings — channels that reach enterprise cloud developers who are unlikely to ever browse the App Store. AWS and Azure each have partner programs (APN, Microsoft Partner Network) that provide co-sell support and listing in their solution catalogues. Revenue share is typically 3–5% on marketplace transactions, but the volume of traffic through these channels justifies the cut.

4.4 Partnership Revenue Summary

Partnership Type	Year 2	Year 3	Year 4	Year 5
Ignition / SCADA ecosystem (referral-driven subs)	\$8,000	\$18,000	\$32,000	\$50,000
Arduino Opta bundle / cross-promo	\$0	\$5,000	\$12,000	\$22,000
Education site licences (TAFE / colleges)	\$12,000	\$30,000	\$60,000	\$100,000
Udemy course bundles	\$3,000	\$10,000	\$20,000	\$35,000
ICS cybersecurity training licences	\$0	\$5,000	\$15,000	\$30,000
Cloud marketplace (AWS/Azure)	\$0	\$2,000	\$8,000	\$20,000
Total Partnership Revenue	\$23,000	\$70,000	\$147,000	\$257,000

👉 Partnership revenue is NOT included in the base-case revenue model at Section 2 (where it is conservatively estimated at \$5K / \$20K / \$45K / \$90K). The figures in this table represent the upside case if partnership outreach is actively pursued. Achieving even 40% of the above would move the base case closer to the optimistic scenario.

5. New Products & Revenue Streams

Beyond feature additions to the existing app, the following products represent distinct, separately-monetisable offerings that leverage the same technical foundation.

5.1 "Industrial Copilot" AI Tier

A premium subscription add-on (\$29.99/month or \$199/year) that embeds a domain-tuned LLM assistant directly in the app — accessible from a slide-over panel while editing Ladder Logic or Structured Text. The assistant understands the current project context (tag names, data types, existing rungs) and can generate code, explain faults, translate natural language to IEC logic, and review programmes for common mistakes. Backed by a fine-tuned model (or a system-prompted GPT-4o / Claude call via API). This is "GitHub Copilot for PLC engineers" — no competing product in the simulator space has it. Target launch: Year 2–3. Expected uptake: 10–15% of active Pro subscribers, generating \$18K–\$50K ARR.

5.2 Mobile Soft PLC Hardware Kit

A companion hardware kit positioned as "bring your simulation to life." Built around an ESP32-S3 or Raspberry Pi Pico W: 8 push-button inputs with debouncing, 8 LED digital outputs, 2 potentiometer analog inputs, 1 small OLED status display, USB-C and WiFi connectivity. The kit pairs with the app wirelessly — pressing a physical button drives the corresponding simulator input tag; a simulated output coil lights the corresponding physical LED. BOM cost at 500-unit volume: approximately \$11–\$16. Target retail: \$49 (direct), \$69 (Amazon/Festo distribution). The hardware is manufactured and fulfilled by a third-party contract manufacturer; zero inventory risk with a print-on-demand model until unit volumes justify a full production run. Creates a hardware revenue stream completely separate from app subscriptions.

5.3 White-Label Protocol Engine Licensing

The in-app OPC UA, Modbus TCP, MQTT, and DNP3 servers are a technically sophisticated piece of engineering — pure Dart, no native binaries, running inside a Flutter app, verified against real industrial clients. Other Flutter-based industrial apps (there are several niche industrial HMI / monitoring apps in the store) would pay for licensed access to this protocol engine rather than building their own. Licensing model: \$5,000–\$20,000 one-time for a source licence, or \$299/month for a compiled plugin package with support. This is a small-volume, high-ACV product — 5 licensees at \$10,000/year each generates \$50,000 ARR with essentially no ongoing cost. Explore Year 3–4 once the protocol stack is OPC Foundation certified.

5.4 Online Training Content Library

A subscription content tier (\$9.99/month "Learner Plan") that bundles guided video walkthroughs, structured exercises, and automated grading — built into the app or delivered via a companion web platform. Content library: 30–50 structured exercises covering LD, ST, FBD, SFC, and the key protocol configurations (how to connect Ignition to Mobile Soft PLC via OPC UA; how to map Modbus registers; how to configure a Sparkplug B topic tree). Produces passive income from learners who are below the "Pro" commitment threshold. Cross-sells easily to Udemy students and TAFE graduates continuing their education. Estimated ARR at 500 learner subscribers: \$60,000/year at very low content refresh cost after initial production.

6. Year-by-Year Execution Roadmap

This roadmap sequences the commercial and technical priorities across five years to maximise revenue growth while remaining executable for a small team.

Year 1 — Launch & Validate

 Revenue Target: \$46,585 gross. Focus: App Store launch quality, community seeding, first 500 paying users.

- **Q1:** Complete Phase 7 (mobile packaging) and Phase 10 (store builds). Publish to Apple App Store and Google Play. Establish Windows direct download via website with Stripe checkout. Set up analytics (RevenueCat for subscription tracking, PostHog for product analytics).
- **Q2:** Begin content marketing — YouTube walkthrough series ("PLC programming on your phone"), Reddit community presence (r/PLC, r/SCADA, r/ControlTheory). Free-tier onboarding optimisation: reduce time-to-first-rung to under 3 minutes.
- **Q3:** Implement Cloud Project Sync (Google Drive / iCloud). Launch PLCopen XML export (basic, enough for community buzz). Join Inductive Automation partner directory — publish OPC UA compatibility post.
- **Q4:** Outreach to top-5 Udemy PLC instructors for course bundle deal. First TAFE campus pilot licence outreach. Review Year-1 analytics: identify top conversion funnels, highest-churn cohorts, most-used protocol.

Year 2 — Scale & Partner

 Revenue Target: \$82,478 gross. Focus: Education partnerships, B2B conversions, AI tier MVP.

- **H1:** Ship Enterprise Multi-Seat Console (MVP). Sign first 5–10 education site licences. Launch S7comm protocol (Siemens European market). Start AI Copilot private beta with Pro subscribers.
- **H2:** Release AI Copilot as paid tier (\$29.99/month). Formalise Arduino cross-promotion agreement. Publish OPC Foundation CTT compatibility test results. Begin BACnet protocol implementation.

Year 3 — Expand Verticals

 Revenue Target: \$142,390 gross. Focus: Cybersecurity vertical, hardware kit, certification pathway.

- Ship BACnet protocol — target building automation integrators via ASHRAE forums and BACnet International
- Launch Hardware Companion Kit (MVP batch of 200 units via Kickstarter-style pre-order to validate demand)
- First ICS cybersecurity training firm site licence deal (target Dragos Academy or similar)
- Submit to OPC Foundation CTT for formal compliance certification
- Explore AWS Marketplace and Azure Marketplace listing for "virtual OT device" positioning

Year 4 — Platform Moves

⚙️ Revenue Target: \$235,435 gross. Focus: API platform, white-label licensing, scale partnerships.

- Launch Virtual PLC API (REST + WebSocket headless engine) — targeted at SCADA software test automation
- White-label protocol engine: first commercial licensee for OPC UA / Modbus stack
- In-app Learning Paths & Certification Mode fully shipped — integration with 3+ college curricula
- 100+ education institution site licences
- Evaluate acquisition interest given \$207K+ net profit run rate

Year 5 — Harvest & Options

🎯 Revenue Target: \$375,829 gross. Options: continue scaling, raise, or sell at ~\$1.73M implied valuation (4.6× ARR).

- All protocol suite complete (EtherNet/IP implicit, S7comm, FINS, SLMP, BACnet)
- 2,400+ paying users across all tiers — substantial enough to attract strategic acquirer interest
- Online Training Library generating passive ARR of \$50K–\$80K with minimal ongoing maintenance
- Decision point: Bootstrapped lifestyle business (\$335K net profit/year), raise seed funding to scale to enterprise, or strategic sale to industrial software consolidator (Aveva, Emerson, Rockwell Software, PTC)

7. Key Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Rockwell / Siemens releases a free mobile companion app	Low-Med	High	Speed to market and depth advantage. Focus on interoperability (PLCopen XML, multiple protocols) — the one thing incumbents are slow to ship.
App Store policy change restricts port binding / background sockets	Low	High	ADR-010 already accounts for iOS foregrounding constraints. WebSocket API fallback mode documented. Monitor Apple developer news.
Low organic discovery in App Store (too niche)	Med	Med	Invest in YouTube/Reddit content, Udemy instructor partnerships, and Inductive Automation listing early — these drive targeted traffic that App Store search cannot.
Education budget cycles slow site licence sales	High	Low-Med	Education revenue grows slowly by design — Y2 target is only 15 institutions. Buffer by over-weighting individual Pro subscriber growth in Y1-2.
AI tier cost exceeds revenue (LLM API costs)	Med	Med	Gate AI features behind per-query or monthly token budgets. Batch expensive operations. Cache common responses. Target \$0.10-\$0.30 per user per day in API cost at \$29.99/month.
Single-developer bus factor	High	High	Document the architecture (already in progress per ARCHITECTURE.md). Prioritise open standards (PLCopen XML, OPC UA) to ensure continuity. Consider hiring a part-time contractor in Year 3.

8. Conclusion & Recommended Next Steps

Mobile Soft PLC Simulator is technically positioned ahead of every direct competitor in its price tier. The core challenge is not product quality — it is distribution and discoverability in a niche that does not naturally browse app stores. The highest-leverage actions are therefore commercial and relational: getting listed where the target audience already looks (Ignition ecosystem, Udemy courses, TAFE curricula) before investing in paid advertising.

The five-year model demonstrates a credible path from \$46K in Year 1 to \$376K gross revenue in Year 5 under base assumptions, with a net profit margin above 88% in mature years. For a single-founder software product with zero marginal delivery cost, this represents an exceptional return on development investment already made.

The three features with the highest individual revenue impact are, in order: (1) PLCOpen XML import/export — removes the top purchase blocker for professional integrators; (2) Enterprise multi-seat console — unlocks institutional bulk purchasing; (3) AI Copilot tier — creates a new price tier and a defensible moat before the window closes. All three are achievable within 18 months of app store launch.

Immediate Next Steps (First 90 Days)

1. Complete Phase 7 (mobile packaging) and Phase 10 (store build pipeline) — get the app live on App Store and Google Play.
2. Register a Windows direct-download page with Stripe payment (paddle.com or Gumroad as simpler alternatives) — do not wait for a Microsoft Store listing.
3. Join the Inductive Automation Integrator Partner directory — fill out the form at inductiveautomation.com/partner/apply. This is free and takes under an hour.
4. Record a 10-minute YouTube demo: "Simulating a PLC on your iPhone — OPC UA connected to Ignition SCADA." This single video will drive the majority of organic qualified installs in Year 1.
5. Start the PLCOpen XML export implementation — even export-only (no import) will generate community buzz in forums (r/PLC, PLCTalk.net, theautomationblog.com).

🏆 The market window is open now. Industrial simulation tools have historically been expensive, Windows-only, and difficult to trial. A mobile-first, protocol-complete, free-to-try tool represents a genuine category shift. First-mover advantage in App Store search rankings and community reputation compounds quickly — the sooner the app is live, the stronger the position against any future competitor.

— End of Report —